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IT FDN 110 GP Au 26: Foundations of Python Programming
Mod 05 – Assignment 05

Advanced Collections and Error Handling

Introduction

In this document I will briefly review the materials learned in this week's module and provide an explanation of my assignment's process in detail.

Dictionaries

This week we move from using 'lists of lists' to using 'lists of Dictionaries'. It was explained how we create each row as a Dictionary and then add these Dictionary rows to a list object. We were also shown how we can go about retrieving data from these objects via their 'keys' as opposed to via their indexes as we did with lists.

Since this 'key' data is stored in the Dictionary object along with the value data itself, the display of the contents of a Dictionary is somewhat less straightforward than doing so with a simple list. We were shown how we should take care to simplify the printing of the contents of a Dictionary to show only the value data (when appropriate) to make it more human-readable.

JSON

A parallel was drawn between Dictionaries and storing data to files in the JSON format in that they both use the key-value pair pattern. JSON stands to JavaScript Object Notation.

Various comparisons were made between JSON and CSV formatting. Positives and negatives were mentioned for both.

We were instructed as to the use of our first 'module' (json) which includes JSON-specific methods for loading JSON data *from* and 'dumping' JSON data *to* files

for persistent storage.

Structured Error Handling

We were introduced to the concept of ‘try’ and ‘finally’ blocks of code that allow us to handle errors and implement code that will still run regardless of less-than-expected outcomes during runtime.

Exceptions

While there are built-in exceptions that can (and will) occur during code execution, we can also raise our own user-defined exceptions to better convey the nature of error given our code’s unique circumstances. We were given specific examples that demonstrate the usefulness of this.

Code Files Management

Stepping away from coding *itself,* we were told about the reasons for and benefits of managing the sharing of our code files in a well-understood manner. While we could simply make our files available to others via a simple network share or even by email, using more sophisticated tools like Github allow us to collaborate while maintaining a history of revisions, backups, code reviews and other benefits.

Intro To GitHub

This week we are beginning to share our code publicly and we are instructed how to create and use a repository in GitHub to do so.

My repository is available at:

https://github.com/defhermit/IntroToProg-Python-Mod05

I chose to use my already existing GitHub account under my username ‘defhermit’. If it would be more appropriate to create and use an account that more closely matches my real name or UW alias, please do let me know!

Assignment Code In English

Here is a pseudocode-list description of the process by which my assignment code undertakes its process:

- Import the json module
- Define the data constants: menu options string and file name string
- Define the data variables: student/class input strings, student data Dictionary, students list to hold Dictionary entries, file reference variable and menu choice input string

Load the existing data from the already-saved file into list using the json module. Handle exceptions that might occur during this process. Close the file when done.

In a continuing loop, display the menu to the user and accept user's decision as to which option to execute (1 through 4).

> If user chooses '1': accept student and course data, load input values into a Dictionary object and append that data to the list object loaded from file. Handle input validation by not allowing numeric characters in students' first or last names.
>
> If user chooses '2': Display current data stored in list of Dictionary entries in human-readable format, omitting 'key' data and including commas as well as 'border' formatting lines
>
> If user chooses '3': save current data stored in list of Dictionary entries to file using the json module's method. Close file when complete. Print the raw data of what is saved to the file. Handle exceptions that might occur during this process.
>
> If user chooses '4': break out of the loop to allow the program to end
>
> If user chooses anything but options 1 through 4, present a message explaining that those are the only valid input options.

At program end, display message indicating that the execution of the program is complete.

Summary

We are once again building on concepts introduced in previous weeks/modules while adding on additional concepts and functionality. This week we adapted our entry adding/display code to utilize JSON data and Dictionaries instead of CSVs and simpler lists. We learned how to handle exceptions that may arise as well as how to raise our own errors/exceptions and how to display customized messages upon those occurrences. We also stepped into the world of source control with an intro to GitHub. I look forward to more work in this regard moving forward!